

# AstraBuild: A Longitudinal Evaluation of GPT-6 Astra as a 3D Engineering Agent

Task Capability, Hypothesis Revision, and Failure Modes in Component-Level Substation Reconstruction

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## Abstract

Can a general-purpose reasoning model do more than generate plausible 3D assets and instead act as an engineering reasoner that *tests and revises modeling hypotheses when physical evidence contradicts them*? We study this question in a longitudinal reconstruction of one operating operating substation. GPT-6 Astra, accessed through Codex, operates inside a fixed Blender/Python harness that supplies registered photogrammetric geometry, field images, engineering records, prior reconstruction state, deterministic tools, and task-specific validators. The preserved record contains **38 sequential core batches across 12 reconstruction task families**; later batches inherit earlier geometry, reusable assets, validators, and failures, so the study is not an IID benchmark.

The record separates two aspects of performance. When a task can be reduced to an explicit measurement problem, the workflow can compile a local reference and delegate numerical estimation to deterministic operators; wall, capacitor, building, and cabin cases then reach centimeter-scale *internal agreement with the same photogrammetric reference used during reconstruction*. The more difficult revisions occur at the hypothesis level. Among seven documented adaptation/failure episodes selected for cross-case analysis, five concern the evaluation protocol, reusable-asset boundary, representation class, or task-selection rule. B15 moves finite geometry out of an overgeneralized shared asset; B23 replaces a straight lead after measured profiles expose roughly 0.5 m of curvature; B25/B26 changes task selection from inventory completion to unexplained geometric coverage; and D38 converts 2D review markup into 3D correction but requires a subsequent review to repair an orientation-semantics error. These cases show that externalized measurements and review artifacts can cause Astra to revise the *kind of engineering hypothesis* being used, while first-pass detail interpretation and coordinate semantics remain error-prone.

Accepted task outputs accumulate into a **37,153-object** station state, providing a long-horizon stress test of reuse, provenance, and composition rather than an accuracy benchmark. The evidence therefore supports a narrower characterization: Astra can participate in a persistent, evidence-grounded 3D engineering process and can revise high-level modeling decisions when contradictions are made explicit, but deterministic validation and human review remain material to acceptance. The present record is limited to one site, has no independent survey reference, and contains no matched reruns with another model.

**Experiment provenance.** The project owner reports the configuration as **Codex + GPT-6 Astra with Extra High reasoning**. The inspected historical Blender batch records do not embed model/reasoning-effort metadata. This setting is therefore marked as user-confirmed rather than independently recovered from the old logs.

## 1 Introduction

Photogrammetry and neural view-synthesis methods can recover impressive scene geometry and appearance [9, 8, 6]. Industrial digital twins, however, often require a different representation: editable devices, reusable component families, explicit connectors and hierarchy, semantic identity, and traceable revision history. A fused surface does not by itself identify which repeated structures should share one master asset, how a transformer inspection point maps to a visible part, or whether a later correction silently invalidated previous work.

The site reconstruction starts from a registered but noisy photogrammetric mesh, UAV/ground imagery, engineering inventories, semantic records, manual position proxies, and human review. These sources play different epistemic roles. Coarse geometry constrains layout and reveals occupied regions; imagery constrains visible component form; inventories constrain expected semantics; human feedback identifies errors or authorizes equivalence; and persistent batch artifacts record what the system previously believed.

General-purpose model work in robotics provides a useful conceptual precedent. Code as Policies and VoxPoser expose executable/spatial interfaces [7, 5]; GPT-Policy separates context construction, fixed-model tool selection, constrained execution, and feedback-driven replanning [3]; and the GPT-6 Astra embodied-policy report studies a System-2 role for semantic reasoning and exception handling [10]. We ask an analogous question for 3D engineering: can a general-purpose model operate a reconstruction process while deterministic geometry code and validators remain the acceptance authority?

This paper is a *longitudinal single-site case study*, not an IID benchmark. Its evidence is therefore a combination of artifact structure, metric ledgers, preserved failures, protocol changes, human-feedback traces, and semantic augmentation rather than one success rate.

## 1.1 Research question

Our central question is:

**When a general-purpose reasoning model is constrained by registered geometric evidence, programmatic tools, and deterministic validation in a persistent 3D environment, how does it form, test, and revise engineering hypotheses, and where do those decisions fail?**

We operationalize this question through four subquestions: which heterogeneous reconstruction tasks the same model/harness can execute; whether contradictions cause parameter changes or revisions to the protocol, abstraction, representation, and task-selection rule; which decisions still require deterministic or human review; and whether accepted outputs remain consistent when composed over a long task horizon.

## 1.2 Contributions

We make three contributions: (1) a longitudinal evaluation protocol in which 12 heterogeneous reconstruction task families probe one focal model through a common evidence/action/validation interface; (2) a cross-case analysis of engineering hypothesis revision using preserved failures and intermediate evidence; and (3) a long-horizon composition stress test that exposes both the benefit and propagation risk of persistent engineering state.

## 2 Related Work

**Surface reconstruction.** Classical structure-from-motion/photogrammetry [9], NeRF [8], and 3D Gaussian Splatting [6] provide increasingly powerful geometry/appearance representations. Our focus is downstream: converting incomplete spatial/image evidence into editable and semantically addressable engineering structure.

**Programmatic 3D and tool-using agents.** Programmatic embodied control [7], 3D language-conditioned objectives [5], and VLA systems [1] demonstrate the value of structured action interfaces. 3D-GPT decomposes procedural 3D generation into task-dispatch, conceptualization and modeling agents [11]; SceneCraft constructs scene graphs, translates spatial relations into Blender-executable code, visually reviews renders and distills reusable functions [4]. AstraBuild differs in problem direction: it reverse-engineers registered physical evidence and therefore relies on metric profiles, route/topology fits, explicit validation domains, history preservation and retained failures. GPT-Policy and the Astra embodied-policy report are particularly relevant in separating context, reasoning, constrained execution and feedback, and in organizing one shared architecture across heterogeneous tasks [3, 10]. RoboDojo illustrates benchmark-style metric breakdowns [2]; AstraBuild borrows the reporting discipline but does not claim an IID benchmark.

## 3 Problem Formulation

Let the persistent reconstruction state be

$$S_t = (M_t, A_t, X_t, Q_t), \quad (1)$$

where  $M_t$  is scene geometry,  $A_t$  reusable asset hierarchy,  $X_t$  semantic/inspection mappings, and  $Q_t$  provenance/validation records. A bounded context is compiled as

$$C_t = \mathcal{C}(G, I, R, U, H_t), \quad (2)$$

from coarse geometry  $G$ , images  $I$ , engineering records  $R$ , manual spatial priors  $U$ , and persistent history  $H_t$ . The model proposes

$$a_t \sim \pi_\theta(\cdot \mid C_t, S_t, f_{t-1}), \quad (3)$$

then deterministic execution and validation produce

$$(\tilde{S}_{t+1}, e_t) = E(a_t, S_t), \quad v_t = V(\tilde{S}_{t+1}, G, Q_{\leq t}). \quad (4)$$

Only artifacts satisfying the relevant batch checks are promoted into the next persistent state; otherwise diagnostic evidence is retained for replanning.

## Can one reasoning agent turn heterogeneous site evidence into a persistent, editable digital twin?

Physical evidence goes in; the same GPT-6 Astra/Codex engineering agent repeatedly proposes programs and revisions; validated component models accumulate into one station state.

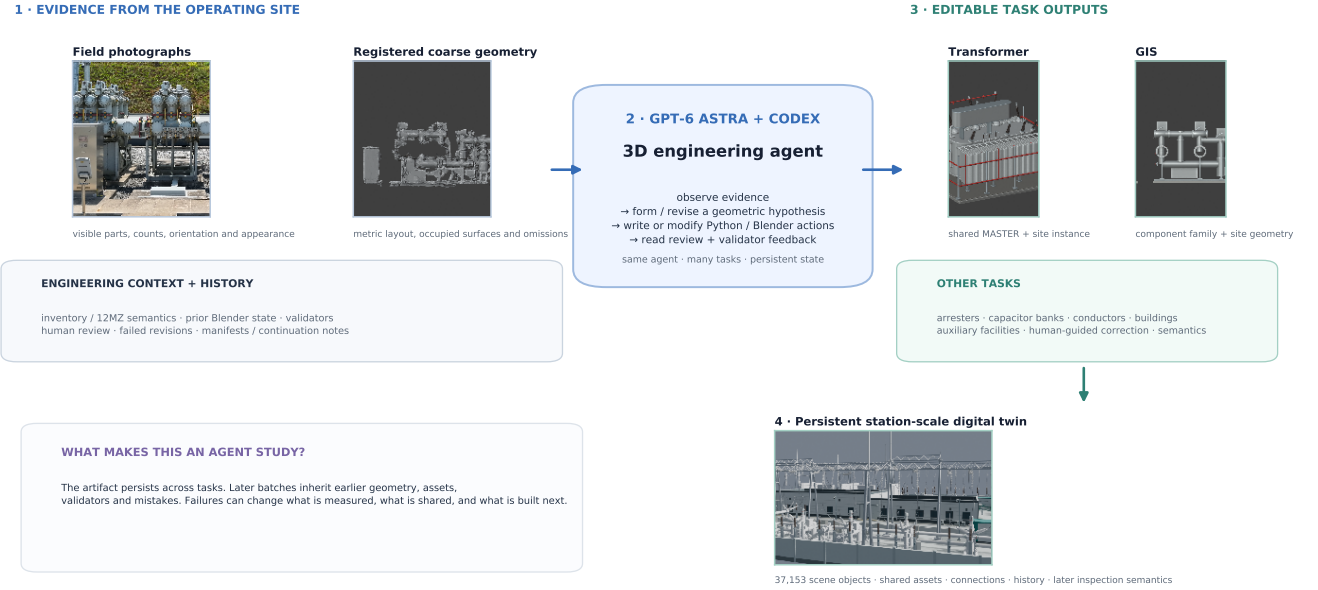


Figure 1 · Study overview. The photogrammetric reconstruction is evidence, not the final representation. AstraBuild targets an editable, componentized and traceable engineering state that accumulates across heterogeneous reconstruction tasks.

**Figure 1:** Study overview. Field imagery, registered coarse geometry, engineering records, and prior history form the evidence context. The same Astra/Codex engineering agent repeatedly proposes and revises programs; validated component models accumulate into one editable station state.

## 4 Study Materials and Provenance

### 4.1 Evidence sources

**Table 1:** Primary evidence sources and their epistemic role.

| Source                       | Recorded scale                                     | Role                                                      | Boundary                                                |
|------------------------------|----------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Photogrammetric mesh         | coarse 23 tiles, $\sim 112 \times 134$ m           | layout, local surface reference, omission discovery       | not independent truth for residuals using the same mesh |
| UAV/ground images            | 775 early identity-photo candidates + later audits | visible form, count, ordering, materials                  | incomplete calibration/identity in several stages       |
| Inventory + 12MZ             | 153 devices, 106 names, 1,842 rows                 | expected semantics, codes, inspection concepts            | inventory presence does not prove geometry completion   |
| UE/manual boxes              | 82                                                 | position prior, cross-check                               | not an independently surveyed reference                 |
| Human review + batch history | 38 core batches + revisions                        | error signals, equivalence authorization, external memory | human intervention is part of the method                |

## 4.2 Version boundary

D38.2 is the headline core-geometry baseline. D40 changes material assignment while validating geometry/transform invariance. D41 is an additive transformer inspection-semantic/accessory layer and the research endpoint. The paper keeps these scopes separate rather than reporting all numbers as one frozen artifact.

# 5 Evaluation Harness and Persistent Engineering State

The evaluation separates the focal model from deterministic execution. GPT-6 Astra selects evidence, proposes or revises an engineering representation, and writes or modifies Blender/Python programs. Numerical fitting, rendering, state preservation, and task-native validators execute outside the language model. A model-authored program is therefore a proposal rather than an accepted result.

## 5.1 Persistent state and evidence context

The formal state/context definitions in Section 3 make this separation explicit. Registered geometry supplies layout and local comparison surfaces; field images supply visible form; engineering records supply expected identity and semantics; human review supplies equivalence authorization or spatial error signals; and prior batch artifacts expose what the workflow previously accepted or rejected.

## 5.2 Model decisions and deterministic execution

Observable Astra decisions include selecting a local reference, choosing a comparison domain, choosing a representation class, constructing shared **MASTER** collections and rigid **SITE** instances, generating finite connectors or conductor paths, requesting diagnostics, and revising programs after review. NumPy/SciPy perform numerical fits, Matplotlib creates profiles and diagnostics, Blender materializes geometry and fixed-camera renders, and validators recompute geometry, contact, continuity, coverage, state-preservation, and file-integrity checks.

This distinction constrains attribution. A small fitted residual shows that the model formulated and invoked a measurement procedure; it does not show that the language model performed numerical optimization internally. Conversely, plausible generated geometry is not accepted because the model produced it. It is promoted to persistent state only after review and task-specific validation.

## 5.3 Review, validation, and external memory

The recurring state transition preserves inherited state, selects a local evidence domain, inspects and plans, builds or revises geometry, renders controlled review evidence, validates task-native properties, and freezes the release and its continuation record. Specialized operators—profile fitting, route diagnostics, coverage audits, collision checks, or markup registration—are invoked only where needed.

Thirty-two formal batches from B07 through D38 retain their original interactive review pages, typically exposing fixed-camera Clean/Overlay/Reference evidence, selectors, sliders, and batch-specific caveats. Output folders additionally preserve profiles, preflight graphics, route diagnostics, validation JSON, manifests, failed revisions, and continuation notes. These artifacts provide the external memory used by later tasks and the evidence base for the case analysis. They do not preserve GPT-6 Astra’s private chain-of-thought; the analysis is therefore limited to the observable engineering decision trail.

## 5.4 Representative task trace: B32 secondary-equipment cabins

B32 separates metric fitting from engineering interpretation. Inherited vertical samples and the station frame are processed by a model-authored measurement program; deterministic rectangular fits produce cabin dimensions, yaw, root transforms, and facade diagnostics. Those measurements establish the cabin envelopes but do not determine the door structure. R1 places one door per cabin and puts the Cabin-A door on the wrong side. Fixed-camera review, low-height registered slices, field photographs, and stair-platform signatures contradict that interpretation. R3 retains one Cabin-A east-side door and two Cabin-B west-side doors and rejects a middle Cabin-B depression previously interpreted as a door.

## One AstraBuild task: evidence → hypothesis → program → review → retained engineering state

B32 reconstructs two secondary-equipment cabins. The figure separates what the agent decides from what deterministic tools compute, render and validate.

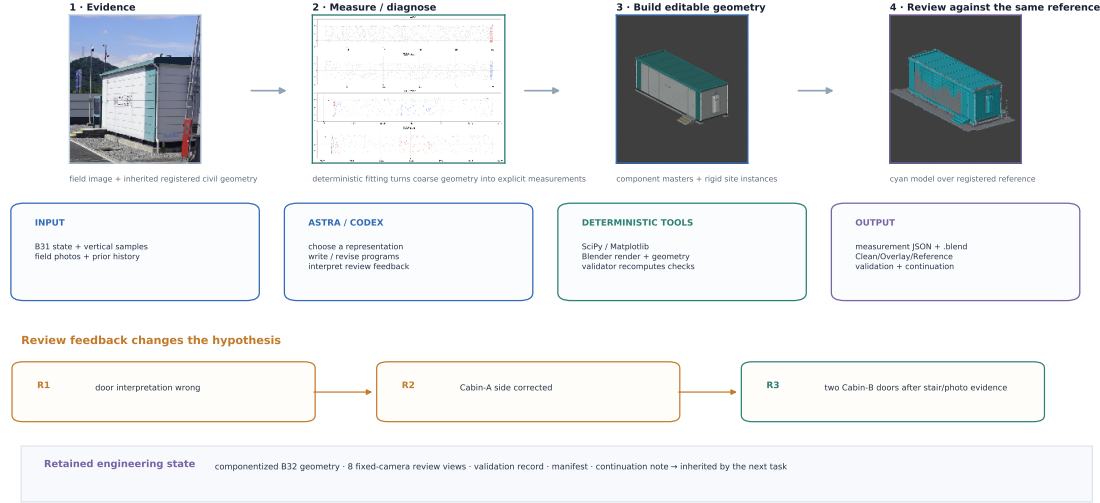


Figure 2 - Observable B32 task trace. Astra/Codex chooses and revises the engineering hypothesis/program; deterministic tools compute measurements, build geometry, render the review, and validate what is allowed into persistent state. The historical record does not preserve private chain-of-thought.

**Figure 2:** Representative B32 task trace. Inherited evidence is transformed into measurements, program revisions, componentized geometry, fixed-camera review, validation, and persistent state; Astra’s engineering decisions are separated from deterministic computation and rendering.

The retained validator also checks inherited-state preservation, facade comparison domains, door/stair constraints, camera framing, and component-library round trips. The original `B32_review.html` remains in the historical record and is exposed by the companion website. B32 therefore illustrates an important boundary for the rest of the paper: a defensible metric envelope does not imply a correct component model at the level of visible engineering detail.

## 6 Evaluation Tasks and Protocol

The evaluation record contains 38 sequential core reconstruction batches spanning 12 task families at one operating substation. The batches are longitudinally dependent: each new task inherits the Blender state, reusable asset library, validation code, and recorded limitations of earlier work. We therefore use the task suite as a set of behavioral probes rather than as IID trials.

The sequence covers surge arresters and local device fitting; walls and gates; main transformers; capacitor banks; VC-A and VC-B GIS families; bus and conductor systems; civil structures and ground; auxiliary facilities; human-triggered correction; a geometry-preserving presentation layer; and inspection-semantic augmentation. The scope also changes over time: early batches emphasize devices and reusable families, middle batches introduce connectivity and coverage-driven omission discovery, and later batches close civil/auxiliary structures and incorporate reviewer feedback.

**Common review protocol.** Thirty-two formal batches from B07 through D38 preserve the original interactive review pages generated during execution. Under fixed review cameras, *Clean* isolates the modeled asset, *Overlay* exposes correspondence with registered coarse geometry, and *Reference* shows the source geometry alone. The companion website exposes these historical pages with their original selectors and split sliders. Quantitative validation remains task-specific because an arrester surface, a conductor route, a civil facade, and an omission-coverage audit do not share a meaningful stationary metric.

**Quantitative context.** The current reference-surface ledger contains 1,550 records with median/p90/p95 distances of 0.036/0.084/0.116 m. These values measure internal agreement with the same photogrammetric mesh that also guides reconstruction; they are not independent survey accuracy. A later 5 cm screening ledger retains 149 passing and 101 non-passing items. In selected high-region coverage audits, the fraction farther than 1 m from modeled geometry changes from 78.8% to 14.6%; this is a regional omission signal, not whole-station completeness.

## Quantitative evidence: agreement, validation screen, and omission coverage

Each panel has a different interpretation; none is an independent survey-accuracy benchmark.

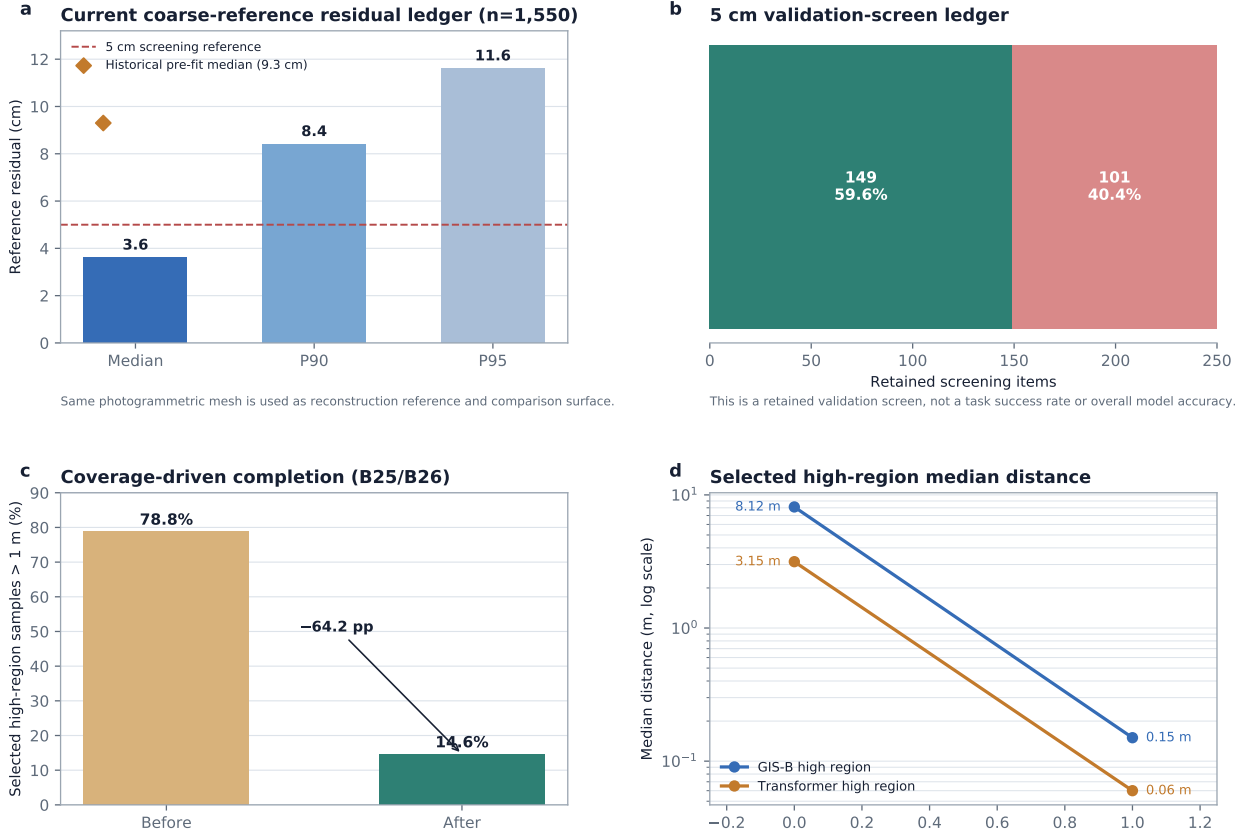


Figure 3 · Quantitative evidence uses separate metrics for local agreement and missing-geometry coverage. The 5 cm screen explicitly retains non-passing items.

**Figure 3:** Task-level quantitative context. Residuals, retained screening items, and coverage-gap measurements answer different questions and are therefore kept separate.

## 7 Behavioral Case Studies: How Astra Revises Modeling Decisions

The task suite establishes what the workflow attempted. The following episodes examine what happens when the current engineering hypothesis is incomplete or contradicted by measurement, review, or inherited state. Cases are selected for diagnostic diversity rather than statistical representativeness.

## Failure and recovery: what changed after the agent was wrong

Each column traces a preserved failure from trigger to protocol change. These are longitudinal case studies, not frequency estimates.

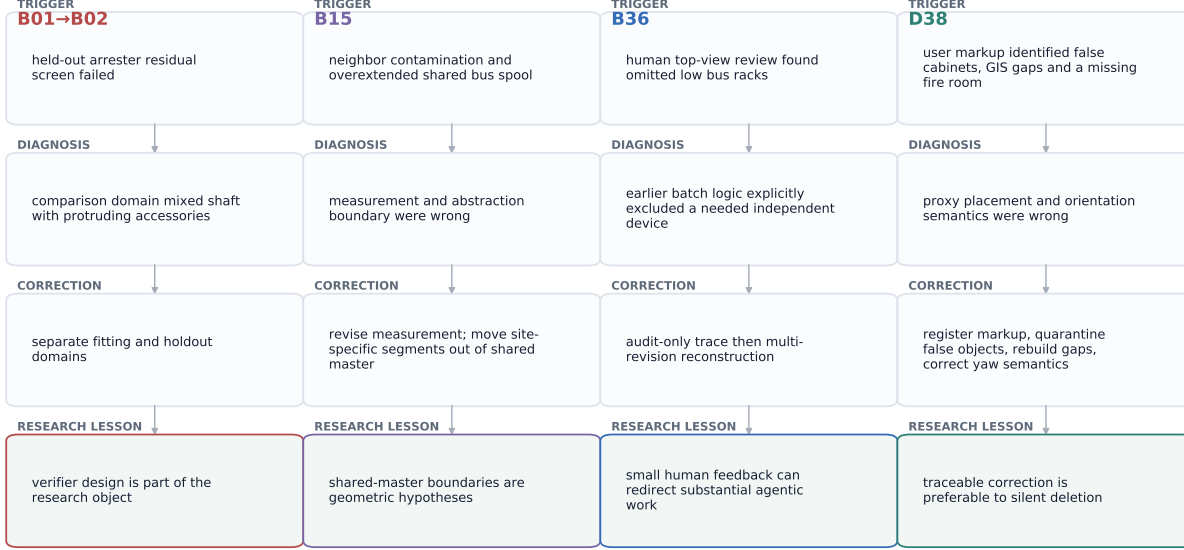


Figure 5 · The central evidence for reliability is not a perfect first attempt; it is whether invalid hypotheses are detected, retained, diagnosed, and converted into explicit protocol or abstraction changes.

**Figure 4:** Representative failure-to-recovery chains. The preserved evidence can force changes to the evaluation protocol, reusable abstraction, geometric representation, or later task selection.

### 7.1 B01 → B02: revise the evaluation domain

Initial arrester installation passed software-level checks but selected held-out RMS values of approximately 0.048/0.073/0.051 m failed a 5 cm screen. The batch was preserved. Inspection showed that the comparison domain mixed intended shaft geometry with protruding accessories. B02 therefore separates fitting and holdout regions and explicitly avoids treating the revised metric as a directly comparable parameter improvement. The protocol, rather than only the fitted dimensions, was part of the hypothesis.

### 7.2 B15: revise the reusable-asset boundary

Neighbor-contaminated measurements and an overextended shared bus-spool assumption required multiple revisions. Finite site-specific geometry had been placed inside a shared master and therefore propagated to installations with different physical extents. The workflow changes both the measurement strategy and the MASTER/site boundary, while archiving obsolete supports rather than overwriting them. Reuse is thus treated as a falsifiable engineering equivalence hypothesis.

### 7.3 B23: change the representation class

An inherited straight-lead hypothesis does not explain the registered neutral-side trajectory, whose mid-span bow is approximately 0.5 m. The workflow extracts measurement profiles, bins observed geometry into control points, fits a curved path with fixed endpoints, and proceeds through retained r1/r2/r3 revisions that add missing soft-lead and contact geometry. Here the evidence changes the class of geometry being built rather than merely a pose or dimension.

### 7.4 B25/B26: change the task-selection rule

Inventory-driven completion cannot expose structures that are absent from the inventory logic itself. The workflow therefore asks a different question: where does the registered coarse geometry contain occupied regions that the component model does not explain? Coverage becomes a task-selection signal. In the selected audited high regions, the fraction farther than 1 m from modeled geometry changes from 78.8% to 14.6%. The result is evidence about those audited regions, not whole-station completeness.

## 7.5 B36: sparse review triggers reconstruction

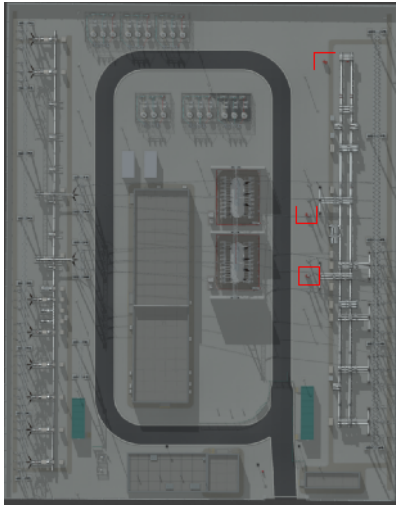
A top-view reviewer identifies missing low transformer-side bus racks. The workflow traces the omission to earlier scope/history, reconstructs the subsystem through multiple revisions, retains an invalid-path draft, corrects a section orientation, and finally moves one segment by 0.16 m to avoid a fire pipe. Human input supplies the spatial error signal; the agent performs the historical search, program revision, reconstruction, and validation.

## 7.6 D38: correction is itself correctable

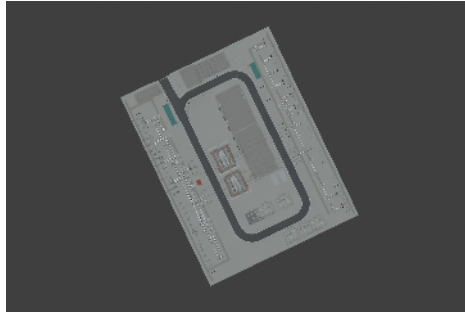
Marked review images are registered to stored views (reported normalized correlations about 0.94/0.92), inverted to model-world coordinates, and used to diagnose three false cabinet groups, three GIS gaps, and missing fire infrastructure. False objects are quarantined rather than silently deleted. A subsequent orientation-semantics error introduced in D38.1 is then caught by overlay review and corrected in D38.2. Both the original error and the correction error remain in history.

### D38 human-in-the-loop correction: markup becomes traceable 3D evidence

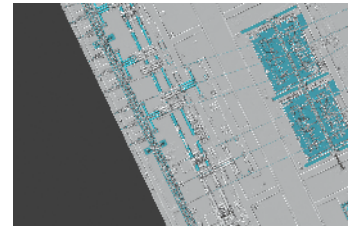
Human review markup (B35 top view)



D38.2 corrected station view



GIS-row correction review



| 1 • Register markup           | 2 • Invert camera mapping                     | 3 • Diagnose                                     | 4 • Preserve provenance                               | 5 • Rebuild + review                                   |
|-------------------------------|-----------------------------------------------|--------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------|
| NCC 0.94 / 0.92<br>scale 0.94 | 7 marked regions →<br>model-world coordinates | 3 false cabinet groups<br>3 GIS gaps + fire room | 9 cabinet objects moved<br>to quarantine, not deleted | GIS gaps + fire facilities;<br>D38.1 yaw error → D38.2 |

Figure 6 • Sparse human feedback supplies an error signal; the agent performs evidence registration, diagnosis, geometric correction, and provenance-preserving quarantine.

**Figure 5:** D38 correction chain. Sparse 2D markup becomes a traceable sequence of registration, coordinate inference, diagnosis, quarantine/rebuild, and re-review.

## 8 Cross-Case Findings

The preserved cases support five descriptive findings about the model-in-harness behavior. The seven adaptation/failure episodes used for layer counts are manually selected documented cases; their distribution is not an estimate of population-wide failure frequency.

**F1: explicit measurement operators separate reasoning from numerical estimation.** Wall, capacitor, building, and B32 cabin tasks reach centimeter-scale internal agreement when the workflow first compiles a registered local comparison domain and then delegates fitting to deterministic code. The visible Astra contribution is selecting

what quantity to estimate, what evidence belongs to the domain, how that domain is split, and which operator is invoked; the numerical optimizer is external.

**F2: difficult revisions occur above parameter tuning.** Among the seven annotated adaptation/failure episodes, five are assigned to protocol, abstraction, representation, or task-selection layers. B01 changes the comparison domain; B15 changes the MASTER/site boundary; B23 changes the representation class; B25/B26 changes how the next task is selected; and D38.1 corrects a coordinate-semantics interpretation. Changing a dimension would not have repaired these underlying assumptions. The five-of-seven count applies only to the selected documented episodes.

**F3: first-pass fine detail and coordinate semantics remain error-prone.** B32 has a defensible cabin envelope while retaining wrong door structure in early revisions; B23 inherits an inadequate straight connection; D38.1 converts a successful image registration into the wrong equipment orientation. In each case, plausible initial geometry is contradicted by an externalized review artifact. Profiles, fixed-camera overlays, and task-specific diagnostics therefore act as part of the measurement process rather than as presentation-only graphics.

**F4: human input is sparse but material.** B08 requires explicit authorization before two transformer assemblies are treated as equivalent; B36 begins from an omission report; D38 begins from marked review images. Human reviewers do not edit mesh vertices, but their interventions change what may be shared, which subsystem is reconstructed, or which region is corrected. The demonstrated workflow is agent-driven and human-steerable rather than autonomous.

**F5: persistent state enables composition and propagates abstraction errors.** Inherited geometry, reusable masters, protected history, and prior validators allow later batches to build on earlier work. The same persistence makes a wrong abstraction consequential: B15 propagates finite geometry through a shared master until the reuse boundary is revised. Provenance, quarantine, and geometry-preserving version boundaries are therefore part of the engineering requirement, not bookkeeping alone.

#### Behavioral analysis of the preserved process record

Statistics mined by scripts/analyze\_process\_behavior.py from the read-only historical batch folders (42 primary installation folders). Revision tags are a lower bound: revisions not frozen under an r-tagged name (e.g. D38.1 → D38.2) are invisible to this scan.

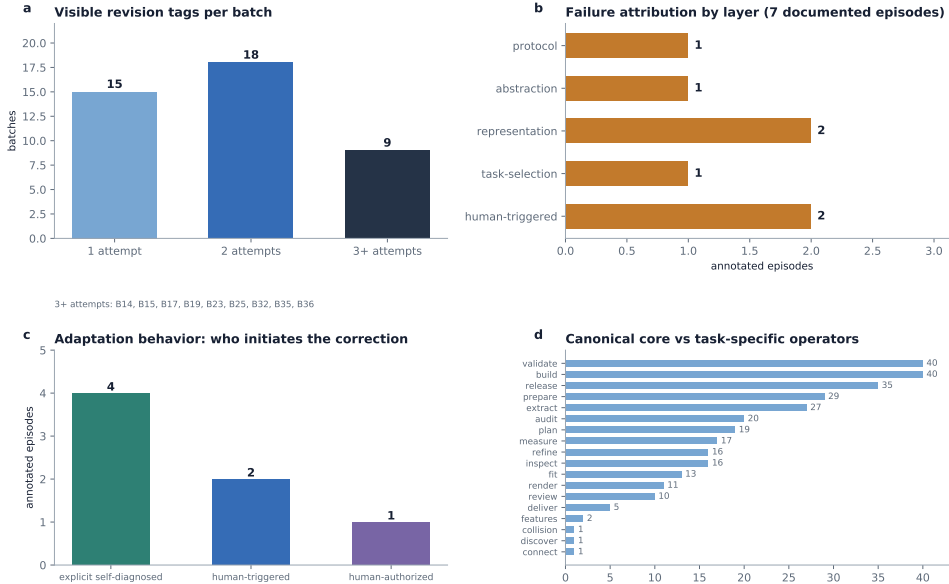


Figure 14 · Panels a-c describe behavior of the reconstruction process (revision, failure attribution, adaptation initiation); panel d shows that build/validate/release form a stable core while measure/inspect/refine/coverage operators activate only on the tasks that need them.

**Figure 6:** Descriptive cross-case context. Visible revision tags are a lower bound; failure-layer and adaptation labels are manual annotations of documented episodes; operator counts describe recorded workflow stages rather than success rates.

## 9 Long-Horizon Composition Stress Test

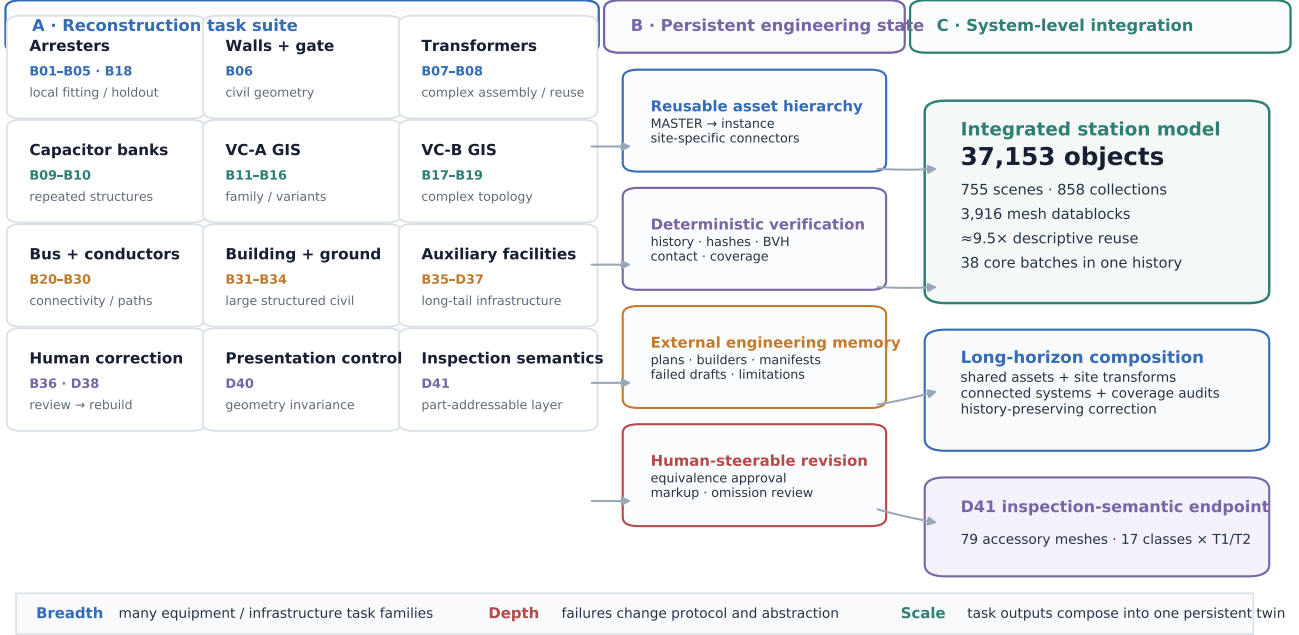
The station-scale state tests whether accepted decisions remain interpretable as subsequent tasks edit the same project. At the D38.2 geometry baseline, outputs from the preceding task sequence coexist in one editable Blender state.

The D38.2/D40-equivalent project contains 37,153 objects, 3,916 mesh datablocks, 755 scenes, 858 collections, and 29,986,649 vertices. These are scale descriptors, not counts of independently verified physical components.

More informative checks concern state discipline: shared masters must remain shared only where equivalence is justified; finite site-specific geometry must remain local; conductor endpoints must continue to meet equipment created in earlier batches; protected transforms and artifacts must survive later edits; and failed variants remain in history. D40 changes presentation materials while validating geometry, transforms, object count, and vertex count as invariant.

## From heterogeneous reconstruction tasks to a persistent station digital twin

The scientific story has three levels: task breadth, behavior under revision, and long-horizon composition into one engineering state.

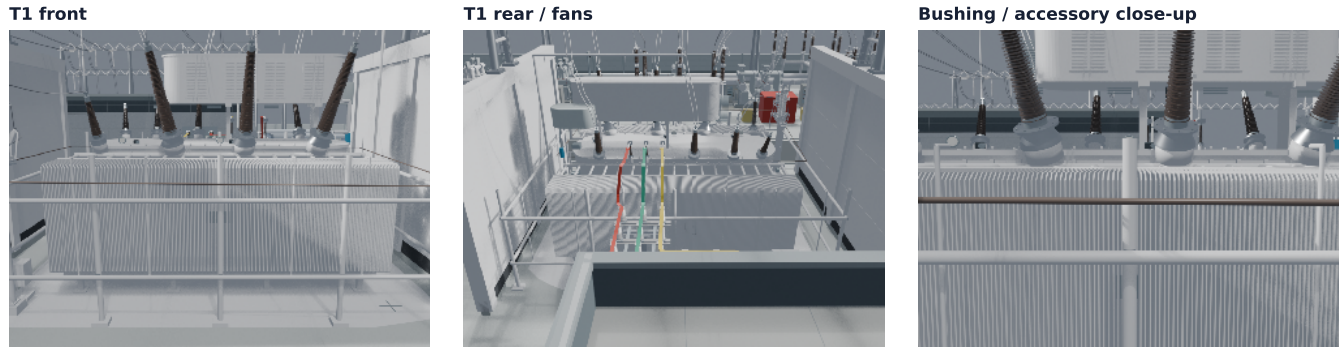


**Figure 7:** Long-horizon composition stress test. Heterogeneous task outputs enter one persistent engineering state, making reuse, provenance, and state consistency constraints on later tasks.

### 9.1 D41 as an additive semantic endpoint

D41 is not a second reconstruction baseline. After the main geometry stabilizes, it adds 79 accessory meshes spanning 17 selected inspection-relevant transformer part classes to a shared accessory collection instantiated on T1 and T2. Validation reports 79/79 selected meshes and 17/17 selected classes on both instances. The result demonstrates one downstream use of the component hierarchy while remaining explicitly separate from the earlier geometry claim.

## D41: augmenting a stabilized transformer with inspection-addressable accessories



|                                                                                               |                                                                                                         |                                                                                               |                                                                                                                        |                                                                                                          |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>79</b><br><b>accessory meshes</b><br><small>selected from semantic source geometry</small> | <b>17</b><br><b>inspection-relevant classes</b><br><small>missing from the B08 visible assembly</small> | <b>2</b><br><b>site instances</b><br><small>T1 and T2 reuse the same accessory master</small> | <b>36</b><br><b>official inspection points</b><br><small>31 model-part + 4 external + 1 non-visual categorized</small> | <b>ok=true</b><br><b>D41 validation</b><br><small>79/79 structure: 17/17 classes per transformer</small> |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|

Figure 7 · D41 is an augmentation layer, not a new station reconstruction. It adds inspection-relevant accessories while reusing the established T1/T2 site transforms.

**Figure 8:** Additive inspection-semantic endpoint. The selected transformer accessory layer is attached after the geometry baseline and is not counted as a new reconstruction baseline.

## 10 Discussion and Limitations

The longitudinal record supports a narrower claim than autonomous reconstruction. Astra repeatedly participates in decisions that determine what is measured, which representation is built, what may be reused, and which unresolved region becomes the next task. Several cases then show the same model making plausible but incorrect decisions at those levels. The practical value of the system therefore comes from coupling model reasoning to external measurements, review artifacts, deterministic validation, and a persistent history that permits later evidence to contradict earlier assumptions.

The closest analogy to the reference embodied-agent evaluations is methodological rather than statistical. Their task suites probe one policy and behavioral episodes explain aggregate outcomes; here, reconstruction tasks probe one persistent model/harness and the episodes explain changes in engineering decisions. The differences are important: these batches are dependent, later tasks inherit earlier state, and validators are heterogeneous because the engineering geometries and failure modes differ.

**Threats to validity.** All task families come from one substation, so diversity within the site does not establish transfer to another site or acquisition protocol. Centimeter-scale residuals compare against the same photogrammetric mesh that guides reconstruction and are not independent survey accuracy. Human steering materially changes B08, B36, D38, and semantic audit decisions. Different task families require different validators, so no defensible global success-rate denominator exists. The seven episodes used for cross-case layer analysis were selected because their evidence trails expose different decision levels; their counts are descriptive, not prevalence estimates. Legacy batch logs do not freeze the model identifier or reasoning effort, which remain owner-confirmed metadata. Finally, the historical record contains no controlled rerun with another foundation model, a specialized 3D system, a no-image/no-coarse ablation, or a professional CAD baseline.

## 11 Conclusion

AstraBuild evaluates GPT-6 Astra through a longitudinal component-level reconstruction record rather than a collection of final assets. Across 12 task families, the model operates through one evidence/action/validation interface while later tasks inherit earlier geometry and mistakes. The central evidence is the sequence of revisions: the workflow changes comparison domains, reuse boundaries, representation classes, and task-selection rules when preserved measurements or review artifacts contradict the current engineering hypothesis.

The record also identifies where the approach remains weak. A good metric envelope does not guarantee correct fine detail; coordinate and orientation semantics can be misinterpreted; human signals materially affect several trajectories; and deterministic validators remain necessary to decide what is retained. The 37,153-object station state shows that accepted outputs can persist and compose over a long horizon, but it is a stress test of state discipline rather than independent evidence of reconstruction accuracy.

The supported conclusion is consequently narrower than “Astra reconstructs a substation autonomously.” The evidence supports GPT-6 Astra as a *human-steerable engineering reasoner inside a deterministic reconstruction system*: useful for selecting, constructing, and revising modeling hypotheses over externalized evidence, but not sufficient as its own acceptance authority.

## A Version Boundary

| Layer | Role                                          | Interpretation                                             |
|-------|-----------------------------------------------|------------------------------------------------------------|
| D38.2 | core station geometry after markup correction | principal geometry baseline for headline artifact counts   |
| D40   | material/presentation layer                   | geometry, transforms and vertex counts checked invariant   |
| D41   | transformer accessory/inspection layer        | additive semantic/geometry augmentation; research endpoint |

## B Validation Taxonomy

The historical report documents recurring checks for history preservation, site transforms, protected hashes, asset round trips, finite geometry, numeric replay, BVH/reference residuals, contact/ground/support continuity, camera framing, rigid-instance SVD, and occupancy/coverage auditing. A future release should encode these checks in a stable versioned schema rather than relying on partially heterogeneous batch JSON structures.

## C Claim Guardrails

- Allowed: “median residual to the same photogrammetric reference mesh is 3.6 cm.” Not allowed: “absolute reconstruction accuracy is 3.6 cm.”
- Allowed: “149/250 retained 5 cm screening items pass.” Not allowed: “the model has 59.6% accuracy.”
- Allowed: “selected high-region > 1 m gap fraction changes 78.8%→14.6%.” Not allowed: “whole-station completeness is 85.4%.”
- Allowed: “the workflow is agent-driven and human-steerable.” Not allowed: “the station was reconstructed fully autonomously.”

## References

- [1] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.
- [2] Tianxing Chen, Yue Chen, Zixuan Li, Junyuan Tang, Kailun Su, Weijie Wan, Baijun Chen, Haoran Lu, Haowen Yan, Honghao Su, et al. RoboDojo: A unified sim-and-real benchmark for comprehensive evaluation of generalist robot manipulation policies. *arXiv preprint arXiv:2607.04434*, 2026.
- [3] Dongzhou Cheng, Taoran Yi, Ye Fang, Xingwu Zhang, Fan Feng, Yixuan Li, Gengxiong Zhuang, Rongze Wang, Shuai Yang, Wei Song, Weizhi Xue, Minyan Wu, Jie Gui, Jiaqi Wang, and Tong Wu. In-context robot learning with vlm agents. Technical report and project website, 2026.

- [4] Ziniu Hu, Ahmet Iscen, Aashi Jain, Thomas Kipf, Yisong Yue, David A. Ross, Cordelia Schmid, and Alireza Fathi. SceneCraft: An llm agent for synthesizing 3d scenes as blender code. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 19252–19282, 2024.
- [5] Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. VoxPoser: Composable 3d value maps for robotic manipulation with language models. In *Conference on Robot Learning*, 2023.
- [6] Bernhard Kerbl, Georgios Kopanas, Thomas Leimk"uhler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023.
- [7] Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control. In *IEEE International Conference on Robotics and Automation*, 2023.
- [8] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *European Conference on Computer Vision*, 2020.
- [9] Johannes L. Sch"onberger and Jan-Michael Frahm. Structure-from-motion revisited. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016.
- [10] Jiayi Su, Yixin Zheng, Mi Yan, Li Yi, Zhizheng Zhang, and He Wang. GPT 6 Astra as an embodied policy. Technical report and code, 2026.
- [11] Chunyi Sun, Junlin Han, Weijian Deng, Xinlong Wang, Zishan Qin, and Stephen Gould. 3D-GPT: Procedural 3d modeling with large language models. *arXiv preprint arXiv:2310.12945*, 2023.